

Results Table — Original vs. Filtered Assembly Statistics

Species: *Cynopterus brachyotis* (lesser short-nosed fruit bat)

Assembly accession: GCA_009793145.1 (ASM979314v1)

Part 2 — Original Genome Assembly Statistics

Metric	Values	Interpretation
Total assembly length	1,758,935,687	1.76 Gb — reasonable for a bat genome (typically 1.7–2.2 Gb).
Number of sequences/scaffolds	48,006	High count suggests a fragmented, draft-level assembly.
Minimum sequence length	1,000	Shortest scaffolds are very small, likely low-information fragments.
Maximum sequence length	4,468,567	4.5 Mb largest scaffold — far below chromosome size, confirming a draft assembly.
Mean/average sequence length	36,639.91	Much lower than the max, showing most sequences are short.
N50	251,278	Moderate contiguity — half the genome is in scaffolds $\geq$ 251 kb.
L50	1,873	Genome is spread across many scaffolds, not a few large ones.
GC content (%)	38.98	Normal for mammals (38–42%); no unusual base composition.

Part 3 — Five Longest Sequences

Rank	Sequence ID	Length (bp)
1	SSHV01002152.1	4,468,567
2	SSHV01001336.1	3,130,546
3	SSHV01001042.1	2,249,068
4	SSHV01002552.1	2,005,528
5	SSHV01001936.1	1,948,660

Part 4 — Original vs. Filtered (≥ 10 kb) Comparison

Metric	Original genome	After ≥ 10 kb filter
Total length	1,758,935,687	1,677,409,341
Number of sequences	48,006	12,483
Maximum length	4,468,567	4,468,567
N50	251,278	267,263
GC content	38.98	38.60

Sequences removed by filter: 35,523 (48,006 – 12,483) **Total length lost:** ~81.5 million bp (~4.6% of original)

Part 5 — ORF Exploration Summary

Metric	Value
Selected sequence	SSHV01000001.1
Sequence length	112,931 bp
Minimum ORF size used	300 bp
Number of ORFs found	40
Longest ORF	1,425 bp (SSHV01000001.1_21)